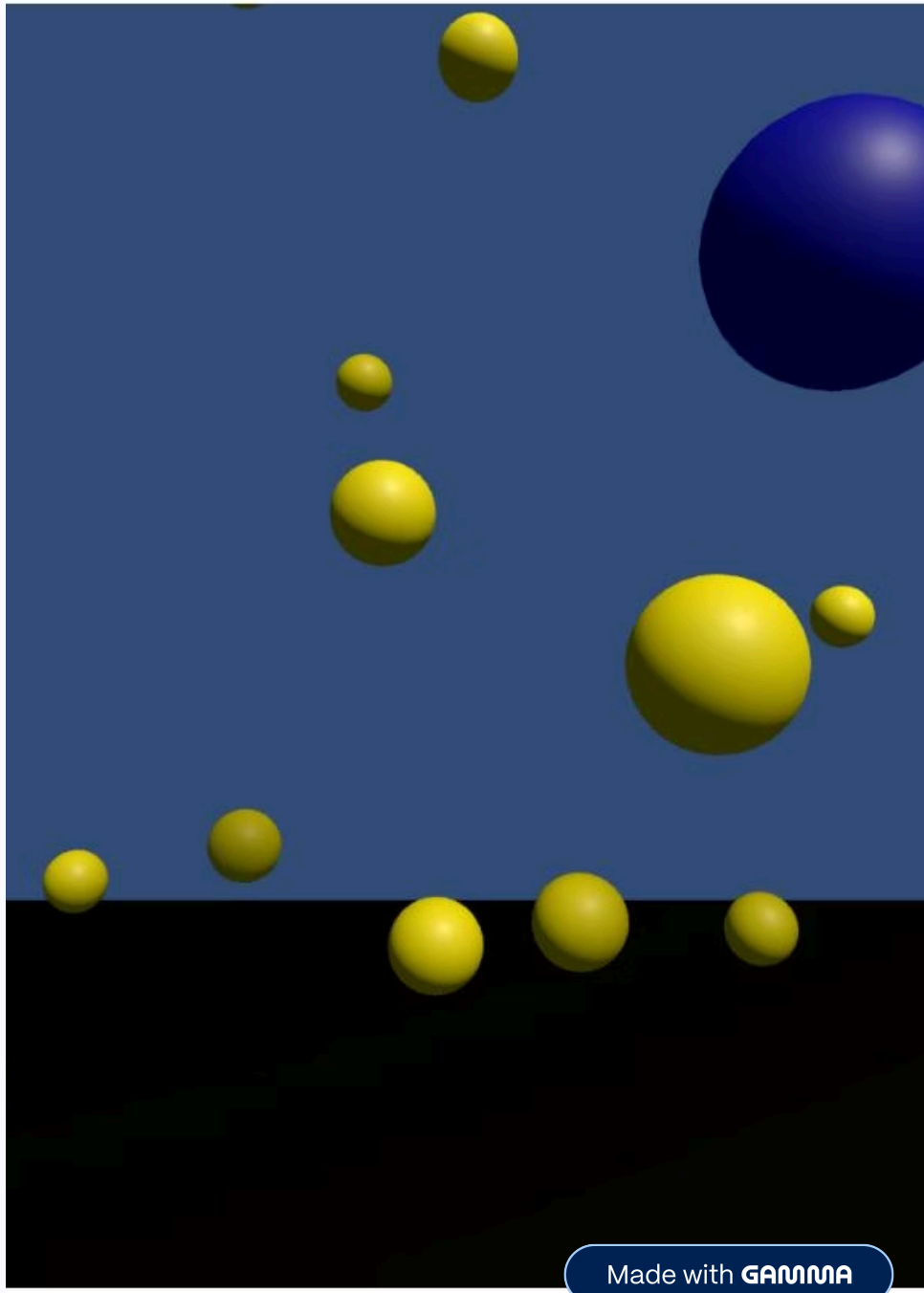
Density-Aware Intent Bubble for VR Selection

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Motivation & Problem

The Problem

- Dense 3D target selection is error-prone and slow
- Direct hand gesture lacks precision in cluttered spaces
- Traditional cursors don't predict user intent in dense environment

Our Approach: WDIB

- Computes intent scores combining spatial distance and hand-motion trajectory alignment
- Predicts which target the user intends to select
- Provides real-time visual feedback

Why This Matters

Core Primitive
Selection is fundamental to VR/AR interaction

Time Cost
Dense layouts increase selection time significantly

High Errors
Frequent re-aiming and wrong target engagement

Research Question: Can intent-aware bubble cursors with predictive green highlighting reduce selection time and improve user comfort in dense VR environments?

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Research Contributions

01

Intent Scoring Algorithm

Lightweight score combining normalized distance and trajectory alignment for real-time prediction

02

Visual Encoding System

Clear color highlighting (green vs. red) that reveals target dynamically based on the score

03

Reproducible Prototype

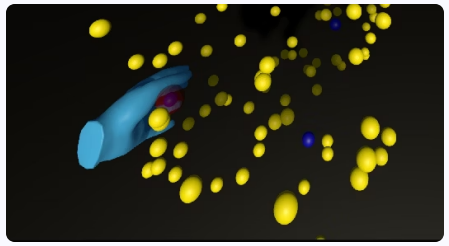
Unity/XR implementation with three experimental conditions for systematic evaluation

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System Overview: Three Conditions

Baseline

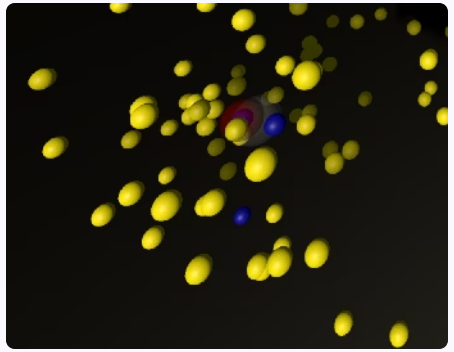
Direct hand movement



Fixed Bubble

Distance-driven visualization

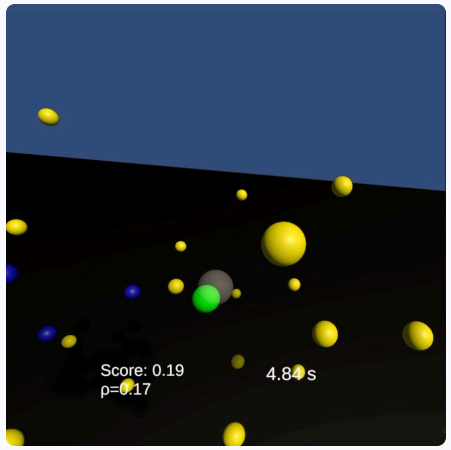
Red highlight



WDIB(Weighted and Density-Aware Intent Bubble)

Intent-score-driven selection

Green highlight



Task setup: Fixed high-density layout with five selectable targets. Bubbles become visible within threshold distance.

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Visual Encoding Strategy

Fixed Bubble

- Shows fixed color visualization
- Distance-based only
- Helpful but not predictive

WDIB (Green)

- Highlights highest-intent target
- Combines intent-score-driven selection with distance and trajectory

Color highlighting feedback provides clear real-time cue for target



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WDIB Model

Overall Scoring Function - 3 parts

$$S_i = w_1(\rho)D_i + w_2(\rho)T_i$$

W1(ρ) and W2(ρ): weight parameters

Si: Score for target i

1. Distance Score

$$D_i = 1 - \frac{d_i}{d_{\max}}$$

Measures how close an object is relative to the farthest candidate

2. Trajectory Score

$$T_i = \max\left(0, \hat{v} \cdot \widehat{p_i}\right)$$

Predicts the user's likely next target based on controller motion

3. Density Term

$$\rho = \frac{N_{\text{near}}}{N_{\max}}$$

Estimates local clutter level and dynamically adjusts weights.

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Hypotheses

1

Selection Time

WDIB reduces selection time vs. baseline hand-motion and fixed bubble approaches

2

Error Reduction

WDIB reduces errors and wrong candidate engagements in dense target layouts

3

Workload

WDIB reduces perceived effort and frustration (NASA-TLX metrics)

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Study Design: Overview

1

Overview

- Within-subjects VR study comparing **Hand Collision**, **Fixed Bubble**, and **WDIB**
- Goal: Evaluate performance, stability, and user experience in **dense selection tasks**

2

Participants

- **N = 5** adult volunteers
- rare to regular VR experience
- All participants used all three techniques (within-subjects)

3

Experimental Conditions

- Randomized layout with:

5 targets + 300 distractors

- High-density environment
- Independent variable: **Interaction Technique**

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Study Design: Task, Measures, and Analysis

1

Task & Procedure

- Task: Select **five targets** as quickly and accurately as possible
- Procedure:
  - Training → Practice → Full trials for all techniques
  - Post-technique subjective ratings (NASA-TLX + preference)

2

Measures

- **Objective:** Completion time, number of attempts
- **Subjective:** Mental/physical demand, stability, frustration, preference

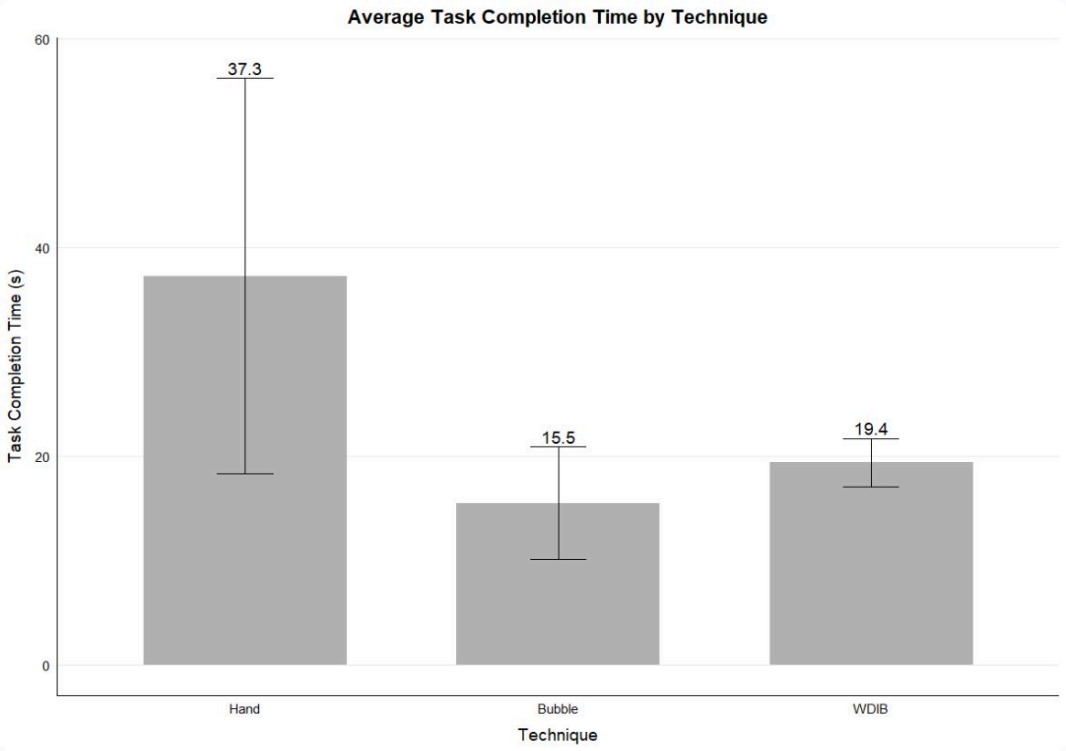
3

Analysis

- Small N → **nonparametric repeated-measures tests**
  - Friedman test for overall effects
  - Wilcoxon signed-rank (Holm correction) for pairwise tests
- **Effect sizes (r)** used to interpret magnitude of differences

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Results:
Objective Performance 1



Task Completion Time

Hand-based selection was also the slowest (M = 37.3 s, SD = 18.9).

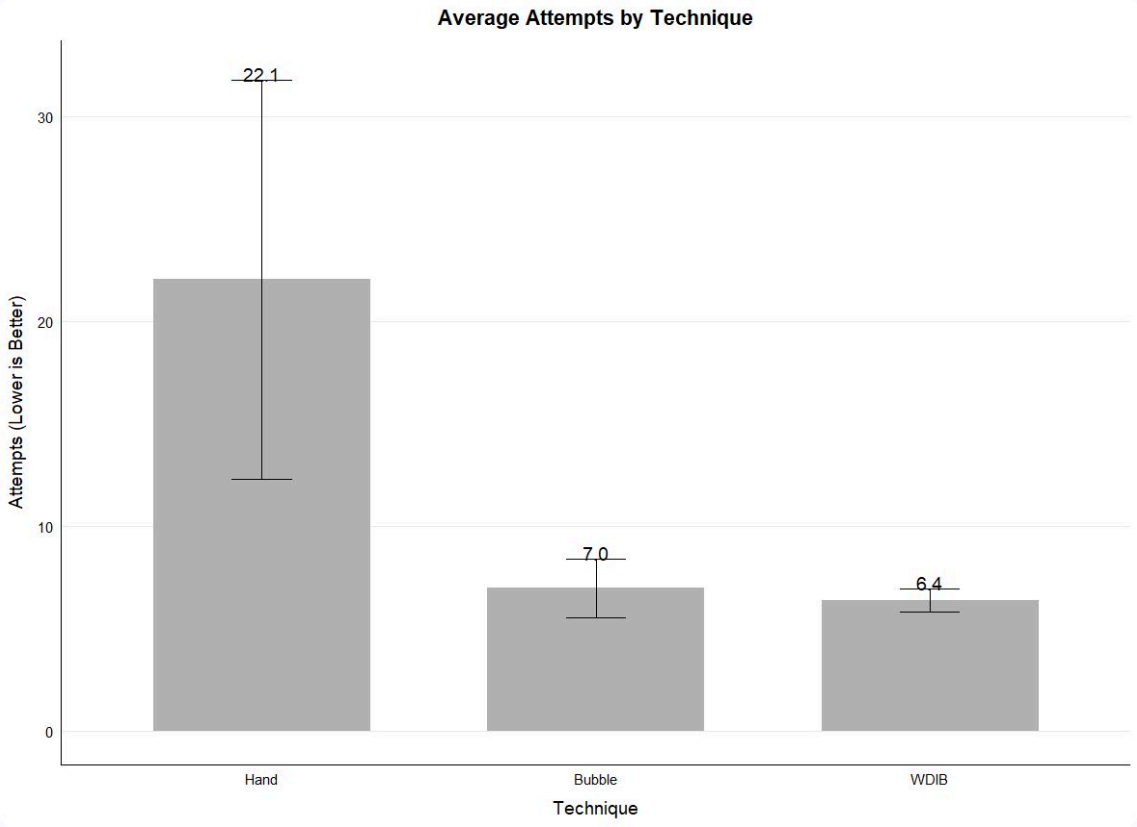
Bubble techniques were substantially faster:

- **Bubble**: M = 15.5 s, SD = 5.4
- **WDIB**: M = 19.4 s, SD = 2.3

Bubble was the fastest overall, while WDIB offered consistent performance with the lowest variability

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Objective Performance 2



Selection Attempts per Target

Hand interaction required the highest number of attempts (M = 22.1, SD = 9.7), indicating frequent accidental activations.

Bubble techniques performed better:

- **Bubble**: M = 7.0, SD = 1.4
- **WDIB**: M = 6.4, SD = 0.6

WDIB showed the fewest attempts, suggesting more stable and predictable selection behavior

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Statistical Analysis

nonparametric repeated-measures tests (Friedman, Wilcoxon with Holm correction) for within-subjects comparisons.

Completion Time - Friedman

No significant overall effect was found ($\chi^2(2) = 2.80, p = .247$), suggesting similar overall performance.

Selection Attempts - Friedman

A **significant overall effect** was observed ($\chi^2(2) = 7.60, p = .022$). Pairwise comparisons did not reach significance due to limited statistical power with N=5.

Robust Effect Sizes - wilcson

- Hand vs. Bubble: **Attempts** $r = .91$, Time $r = .78$
- Hand vs. WDIB: **Attempts** $r = .91$, Time $r = .78$
- Bubble vs. WDIB: Attempts $r = .49$ (moderate)

These larger effect sizes highlight meaningful differences despite the small sample.

Interpretation

Bubble, WDIB showed **consistently large advantages over Hand-based selection**. WDIB performed comparably to Bubble, with indications of slightly fewer attempts and more stable behavior.

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Subjective Ratings and User Feedback

WDIB
Highly preferred in crowded scenes (80% of participants). Reported as more stable (M = 5.6), less physically demanding (M = 1.8), and generally “predictive” and “less tiring.”

Fixed Bubble
Rated as the least mentally demanding, though slightly less stable than WDIB. Preferred by one participant.

Hand-based
Rated as the most frustrating and physically demanding. Often described as “hard to control.”

Overall
Participants showed a strong preference for WDIB due to its stability, predictability, and reduced effort in complex VR environments.

Table 1: Mean subjective ratings (1 = low, 7 = high).

Technique	Stability	Mental	Physical	Frustration
Hand	2.8	4.4	4.8	4.3
Fixed Bubble	5.2	1.5	2.2	2.4
WDIB	5.6	3.4	1.8	1.6

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Limitations

Residual Ambiguity

Despite intent scoring, ambiguity persists when multiple targets have overlapping positions.

Hand-Bubble Size Mismatch

The virtual hand model was disproportionately large compared to the bubble cursor, affecting spatial perception and selection accuracy.

Late Visual Feedback

Color-coded recommendations only appear when the bubble cursor is close to targets, limiting early guidance during approach.

Non-Uniform Spatial Distribution

Random object spawning resulted in non-uniform spatial distributions across trials, introducing variability in task difficulty.

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Conclusion

Key Takeaways

- **Intent-aware selection improves reliability.**
  Combining distance and trajectory cues reduces accidental activations and stabilizes performance in dense scenes.
- **WDIB enhances user experience.**
  Adaptive scoring and real-time visual feedback make WDIB feel more predictable and less tiring.
- **Volumetric techniques outperform hand-based input.**
  Both Bubble and WDIB were faster and required fewer attempts than direct hand collision.

Design Implications

- **Predictive assistance for VR interfaces.**
  WDIB principles can support menu selection, tool interaction, and object-heavy environments.
- **Adaptive selection volumes.**
  Dynamically adjusting selection regions can reduce mental and physical effort.
- **Path toward multimodal intent estimation.**
  Future systems could combine motion, gaze, and gesture for more robust prediction.

Moving from distance-only cues to intent-aware selection is a meaningful step forward for VR interaction, with potential applications beyond object selection into navigation and UI control.

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Team Member Contributions

Yongyun

- Built the selection environment from scratch, including random object and target spawning.
- Implemented the Fixed Bubble Cursor and hand-controller interaction mechanics.
- Developed the distance-based bubble visibility system.
- Created the selection timing system and data logging pipeline.

Daniel

- Designed and implemented the WDIB intent scoring algorithm.
- Developed the green color-coded prediction feedback for highest-intent targets.
- Designed, deployed, and integrated the NASA-TLX questionnaire for subjective evaluation.

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